EE2025: Engineering Electromagnetics

Tutorial 5: Waveguides

July-Nov 2025

- Use appropriate approximations and work with reasonable assumptions wherever necessary.
- Good familiarity with Maxwell's equations helps.

Waveguides

1. In this course, we have seen the phenomena of dispersion occur in two separate occasions, first, with rain drops, and second, with waveguides. Are the *physical* reasons for dispersion the same in both cases? Elaborate. What parameters control the amount of dispersion in both cases?

   In a parallel plate waveguide, which modes shows dispersion, and which one is non-dispersive and why?

Solution:

- The physical reason of dispersion are not same for the both cases:
  Case 1 - Raindrops:

  The anomalous dispersion equation for a medium can be expressed as:

$$n(\omega) = 1 + \frac{q^2 e^2}{\epsilon_0 m} \sum_k N_k \frac{\omega_k^2}{\omega^2 - \omega_k^2 + i\gamma_k \omega}$$

  where: - n is the refractive index, - q is the charge of the electrons, - e is the elementary charge, - ϵ_0 is the permittivity of free space, - m is the mass of the electrons, - N_k is the number of electrons per unit volume, - ω_k is the natural frequency of the k-th oscillator, - γ_k is the damping factor of the k-th oscillator.

  Since the refractive index $n(\omega)$ is frequency dependent, the velocity will also depend on frequency. For raindrops, the medium (refractive index $n(\omega)$) causes dispersion, meaning that different frequency components travel at different velocities.

  Case 2 - Waveguides:

  In a rectangular waveguide, the group velocity v_g can be expressed as:

$$v_g = \frac{v_1}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}}$$

  where f_c is the cutoff frequency, and f is the operating frequency.

  For the m-th mode (TE or TM) in a rectangular waveguide with dimensions a and b, the cutoff frequency $f_{m,n}$ is given by:

$$f_{m,n} = \frac{v_1}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

  .

  In a waveguide, the physical parameters the dimension a and b of the waveguide—affect the propagation of waves. From equation of group velocity, we can see that the there is variations in group velocity by changing the dimension of the waveguides, even when the medium filling the waveguide is non-dispersive ϵ_r, not dependent on frequency.

- In parallel plate waveguide, the group and phase velocity of TE or TM can be written as.

$$v_g = \frac{v_1}{\sqrt{1 - \left(\frac{mc}{2df}\right)^2}}$$

and phase velocity is ;

$$v_p = v_1\sqrt{1 - \left(\frac{mc}{2df}\right)^2}$$

At a given frequency different modes travel with distinct phase and group velocities and also they travel with different phase/group velocity with different frequency. The TE_0 mode does not exist, Other modes of TE travels shows the dispersion.

For the TM_0 mode, the electric and magnetic fields exist like TEM modes. In this case, the phase and group velocity, v_1, which is independent of frequency, making the TM_0 mode non-dispersive in nature. Other TM modes will dispersive in nature.

Reference: RKS: page no- 272-273 (topic: phase velocity and dispersion)

2. A parallel-plates waveguide has a height of 2 cm and is filled with a dielectric of relative permittivity 2.25. The wave propagates in the z-direction as shown below. The magnetic field inside the waveguide is characterized by

$$H_z = 50\cos(100\pi x)\cos\left(3\pi \times 10^{10}t - \beta z\right)\ A/m$$

Determine:

1. The mode that is propagating
2. The Phase constant

Solution:

1. Since the wave is propagating in z direction and the component of H in z direction is non-zero, the mode that is propagated is TE.
   Mode number can be obtained as $\frac{m\pi}{d} = 100\pi \implies m = 2$
   Hence the mode that is propagated is TE_2
2. $k = k_1\sqrt{1 - (\frac{m\lambda}{2d})^2} = \frac{2n\pi f}{c}\sqrt{1 - (\frac{mc}{2ndf})^2} = \frac{3\pi\times10^{10}\sqrt{2.25}}{3\times10^8} \times \sqrt{1 - (\frac{2\times3\times10^8}{2\times\sqrt{2.25}\times0.02\times1.5\times10^{10}})^2}$
   $= 351.24$ rad/m

3. Consider a rectangular waveguide with dimensions 2.28 cm × 1.01 cm. What modes will propagate in this wave guide, if the driving frequency is 17 GHz ? Suppose you wanted to excite only one TE mode; what range of frequencies could we use? What are the corresponding wavelengths (in free space)?

Solution:

$$f_{10} = \frac{c}{2a} = 6.6\ GHz$$

$$f_{20} = \frac{2c}{2a} = 13.2\ GHz$$

$$f_{30} = \frac{3c}{2a} = 19.7\ GHz$$

$$f_{01} = \frac{c}{2b} = 14.9\ GHz$$

$$f_{02} = \frac{2c}{2b} = 29.7\ GHz$$

$$f_{11} = \frac{c}{2}\sqrt{\frac{1}{a^2} + \frac{1}{b^2}} = 16.2\ GHz$$

Four modes occurs f_{10} , f_{20}, f_{01} and f_{11} when the driving frequency is 17 GHz. To get one mode (f_{10}), the frequencies should be between f_{10} = (6.6 GHz) and mode f_{20} = (13.2 GHz). The corresponding λ are between 2.28 cm to 4.56 cm.

4. An air-filled rectangular waveguide operates at 1 GHz and has $a = 5cm$ and $b = 2cm$. Show that the TM_{21} mode cannot propagate at this frequency.

Solution:

For TM_{21}, $f_c = \frac{1}{2\sqrt{\mu\epsilon}}\sqrt{(\frac{2}{a})^2 + (\frac{1}{b})^2} = 9.604GHz$.

Since $1GHz < f_c$, it does not propagate.

5. Calculate values of β, v_p, λ_g for a 2.5 cm by 1.5 cm rectangular waveguide operating at 7.5 GHz, if the waveguide is air-filled.

Solution:

- $\lambda = \frac{c}{f} = 4cm.$
- Let $F = \sqrt{1 - (\frac{\lambda}{2a})^2} = 0.6$
- $\lambda_g = \frac{\lambda}{F_1} = 6.7cm$
- $\beta = \frac{2\pi}{\lambda_g} = 94.2radm^{-1}$
- $v_p = 5 \times 10^8 ms^{-1}$

6. A section of a rectangular waveguide of cross-section 2 cm $\times$ 1.5 cm is to be used as a delay line in a radar at 10 GHz. What should be the length of the section to realize a delay of 10 nsec ?

Solution:

Cutoff frequency of TE_{10},

$$f_c = \frac{c}{2a} = \frac{3 \times 10^8}{2 \times 2 \times 10^{-2}} = 7.5\ \text{GHz}$$

Cutoff frequency of the next higher order mode TE_{01} (Since, $a < 2b$ can't be TE_{20}),

$$f_c^{'} = \frac{c}{2b} = \frac{3 \times 10^8}{2 \times 1.5 \times 10^{-2}} = 10GHz$$

Thus only possible mode of transmission is TE_{10}

Group velocity

$$v_g = c \times \sqrt{1 - \left(\frac{f_c}{f_o}\right)^2} = 3 \times 10^8 \times \sqrt{1 - \left(\frac{7.5}{10}\right)^2} = 1.984 \times 10^8 ms^{-1}$$

Length for a delay of 10 ns,

$$l = v_g \times 10 \times 10^{-9} = 1.984m = 198.4cm$$

7. An air-filled section of an X-band (8.2–12.4 GHz) rectangular waveguide of length l is used as a delay line. The longer dimension of the waveguide is a cm and it operates at TE_{20} mode. Given that, $l = 50$ cm provides a delay of 2 ns at 10 GHz, find a.

Solution: Given the length l and delay τ, group velocity can be calculated as:

$$\nu_g = \frac{l}{\tau} = 2.5 \times 10^8 \text{ m/s} \tag{1}$$

The group velocity is expressed as:

$$\nu_g = c\sqrt{1 - \left(\frac{f_c}{f}\right)^2} \tag{2}$$

where, c is velocity of light in free space, f_c is the cut-off frequency and f is the frequency of operation. The cutoff frequency obtained will be $f_c = 5.52GHz$.

The waveguide dimension a for TE_{20} mode can be obtained as:

$$a = \frac{c}{f_c} = 5.4 \text{ cm} \tag{3}$$

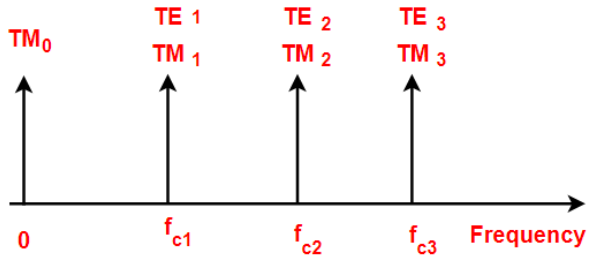
8. A parallel plate waveguide has a separation $d = 1$ cm, and is filled with teflon, having dielectric constant $\epsilon_r = 2.1$. Determine the maximum operating frequency such that only TEM mode will propagate. Also find the range of frequencies over which TE_1 and TM_1 (m=1) modes will propagate, and no higher order modes.

Solution:

The cutoff frequency for the first waveguide mode:

$$f_{c_1} = \frac{mc}{2\sqrt{\epsilon_r}d} = \frac{1 \times 2.99 \times 10^{10}}{2\sqrt{2.1} \times 1} = 1.03 \times 10^{10} \text{ Hz} = 10.3 \text{ GHz}$$

To propagate only TEM waves, we must have $f < 10.3$ GHz.



Cut-off frequencies for different TE and TM modes

To allow TE_1 and TM_1 (along with TEM) only, the frequency range must be $f_{c_1} < f < f_{c_2}$, where $f_{c_1} = 2f_{c_2}$.

Thus the frequencies at which we will have m = 1 modes and TEM will be 10.3 GHZ $< f <$ 20.6 GHz.

9. In the previous question, if the operating frequency is 150 GHz, how many waveguide modes will propagate?

Solution: For mode m to propagate , the requirement is $\lambda < \lambda_c$. 150 GHz corresponds to 2 mm. For the given waveguide and wavelength, the inequality becomes

$$\lambda_c = \frac{2\sqrt{\epsilon_r}d}{m}$$

$$2 \text{ mm} < \frac{2\sqrt{2.1}(10 \text{ mm})}{m}$$

From which,

$$m < \frac{2\sqrt{2.1}(10 \text{ mm})}{2mm} = 14.5$$

Thus the guide will support modes at the given wavelength upto order of **m = 14**. Since there are TE and TM mode for each value of m, this gives, in addition to TEM mode, a total of **29** modes that are above cut off

10. In the waveguide mentioned in the previous question, if the operating frequency is 25 GHz, then the modes m = 1 and m = 2 are the only allowed modes (apart from TEM). Determine the group delay difference (the difference in arrival times between different modes) between these two modes over a distance of 1 cm.

    Technically, even the TEM mode is a propagating mode. Find out which pair of modes has the largest group delay over the same distance.

Solution:

The group delay difference is expressed as

$$\Delta t = \left(\frac{1}{v_{g2}} - \frac{1}{v_{g1}}\right) (\text{s/cm})$$

We have relation for group velocity,

$$v_{gm} = \frac{c}{n}\sqrt{1 - \left(\frac{\omega_{cm}}{\omega}\right)^2} = \frac{c}{n}\sin\theta_m$$

along with the results of question 1 and question 2, we have

$$v_{g1} = \frac{c}{\sqrt{2.1}}\sqrt{1 - \left(\frac{10.3}{25}\right)^2} = 0.63c$$

Likewise,

$$v_{g2} = \frac{c}{\sqrt{2.1}}\sqrt{1 - \left(\frac{20.6}{25}\right)^2} = 0.39c$$

$$\Delta t = \frac{1}{c}\left[\frac{1}{0.39} - \frac{1}{0.63}\right] = 3.3 \times 10^{-11} \text{ s/cm} = 33 \text{ ps/cm}$$

This computation gives a rough measure of modal dispersion in the guide.

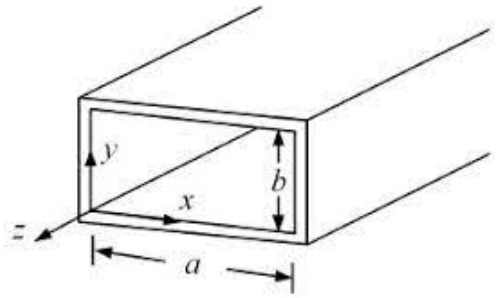
If we consider the delay between TEM and m = 2 mode

$$\Delta t_{\text{net}} = \frac{1}{c}\left[\frac{1}{0.39} - 1\right] = 52 \text{ ps/cm}$$

11. For the fundamental mode inside a rectangular waveguide the logitudinal magnetic field is given as

$$H_z = 20\cos(10y)e^{-j\beta z}A/m$$

Find the cut-off frequency of the mode. Also, find the frequency at which the group velocity is 1/3 of the phase velocity.



Solution:

Given, $H_z = 20\cos(10y)e^{-j\beta z}A/m$

It is in TE_{01} mode. Then, $H_z = C\cos\left(\frac{\pi}{b}y\right)e^{-j\beta z}A/m$

$C = 20$, $\frac{\pi}{b} = 10$

cut-off frequency, $f_c = \frac{\left(\frac{\pi}{b}\right)}{2\pi\sqrt{\mu_0\epsilon_0}} = \frac{10}{2\pi\sqrt{\mu_0\epsilon_0}} = \frac{10}{2\pi}c = 0.478GHz$

Given, group velocity is 1/3 of the phase velocity i.e. $v_g = \frac{1}{3}v_p$

As, $\sqrt{v_p v_g} = c \Rightarrow v_p = \sqrt{3}c \Rightarrow \frac{c}{\sqrt{1-(\frac{f_c}{f})^2}} = \sqrt{3}c$

$f = \sqrt{\frac{3}{2}} f_c = 0.585GHz$

12. For a perpendicularly-polarized wave getting reflected from a PEC (perfect electric conductor) at $x = 0$, if another PEC is placed at a distance $x = d$ such that that tangential electric field goes to zero, wave propagates and it is a parallel plane waveguide. Suppose, instead of a PEC, what if we place a PMC (perfect magnetic conductor) at $x = d$. Obtain the value(s) of d for which the wave propagates. (Hint: Analogous to PEC, the tangential magnetic fields are zero on PMC).

Solution: The magnetic field for TE wave getting reflected from a PEC is given as:

$$\bar{H} = 2j\frac{E_{i0}}{\eta_1}\sin(\beta_1 x\cos\theta)e^{-j\beta_1 z\sin\theta}\cos\theta\hat{x} + 2\frac{E_{i0}}{\eta_1}\cos(\beta_1 x\cos\theta)e^{-j\beta_1 z\sin\theta}\sin\theta\hat{z} \tag{4}$$

For H_z to be zero at $x = d$,

$$\beta_1 x\cos\theta = (2n+1)\frac{\pi}{2} \tag{5}$$

Therefore,

$$x = d = \frac{(2n+1)\lambda_1}{4\cos\theta} \tag{6}$$

13. A broad class of non-magnetic dispersive media exhibit relative permittivity, ϵ_r, that varies as the square of frequency over a narrow range of frequencies centered at ω_0. For these media we presume

$$\epsilon_r = K\left(\frac{\omega}{\omega_0}\right)^2,$$

where K is a real valued positive constant. What is the phase and group velocity for a plane wave in this material?

Solution:

First, note

$$\beta = \omega\sqrt{\mu_0\epsilon} = \omega\sqrt{\mu_0\epsilon_0}\sqrt{\epsilon_r} = \frac{\omega^2\sqrt{K}}{\omega_0}\sqrt{\mu_0\epsilon_0}$$

The phase velocity is:

$$v_p = \frac{\omega}{\beta} = \frac{\omega_0}{\sqrt{K}\cdot\omega\sqrt{\mu_0\epsilon_0}}$$

Whereas the group velocity is:

$$\begin{aligned} v_g = \frac{\partial\omega}{\partial\beta} &= \left(\frac{\partial\beta}{\partial\omega}\right)^{-1} \\ &= \left(\frac{\partial}{\partial\omega}\sqrt{K}\cdot\frac{\omega^2}{\omega_0}\sqrt{\mu_0\epsilon_0}\right)^{-1} \\ &= \left(2\sqrt{K}\cdot\frac{\omega}{\omega_0}\sqrt{\mu_0\epsilon_0}\right)^{-1} \end{aligned}$$

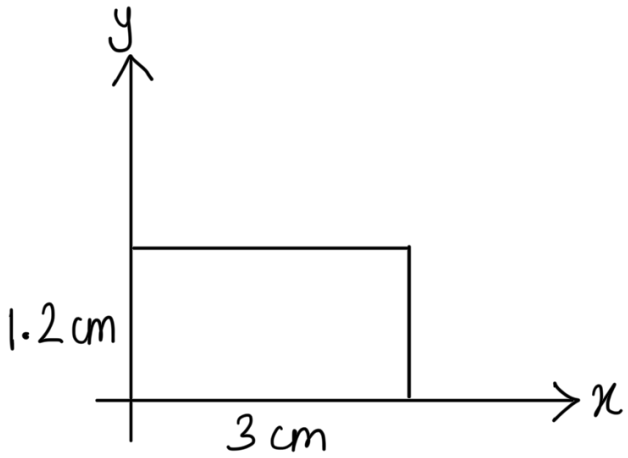
On simplifying, we get

$$v_g = \left(\frac{2\beta}{\omega}\right)^{-1} = \frac{1}{2}\frac{\omega}{\beta} = \frac{1}{2}v_p$$

Thus, we see that in this case the group velocity is always half the phase velocity.

14. The magnetic field along the propagation direction inside a air-filled rectangular waveguide with the cross section shown in the figure is given by:

$$H_z = 3\cos\left(2.094 \times 10^2 x\right)\cos\left(2.618 \times 10^2 y\right)\cos\left(6.283 \times 10^{10} t - \beta z\right)$$



Reason out whether the H field given is feasible for this waveguide.

Solution:

For the rectangular waveguide as per the given figure with dimensions $a = 3\,\text{cm}$ and $b = 1.2\,\text{cm}$, the magnetic field is given by:

$$H_z = 3\cos\left(2.094 \times 10^2 x\right)\cos\left(2.618 \times 10^2 y\right)\cos\left(6.283 \times 10^{10} t - \beta z\right)$$

From the above equation, we have:

$$\omega = 6.283 \times 10^{10}\,\text{rad/sec}$$

The frequency f is given by:

$$f = \frac{\omega}{2\pi} = \frac{6.283 \times 10^{10}}{2\pi} = 10\,\text{GHz}$$

we know,

$$\frac{m\pi}{a} = 2.094 \times 10^2 \quad \text{or} \quad \frac{m}{a} = 66.65\ /\text{m}$$

also,

$$\frac{n\pi}{b} = 2.618 \times 10^2 \quad \text{or} \quad \frac{n}{b} = 83.33\ /\text{m}$$

The cutoff frequency f_c is given by:

$$f_c = \frac{c}{2}\sqrt{(m/a)^2 + (n/b)^2} = \frac{3 \times 10^8}{2}\sqrt{(66.65)^2 + (83.33)^2} = 16\,\text{GHz}$$

The wave with a frequency of 10 GHz will not propagate through the waveguide. Thus, the phase velocity of the wave inside the waveguide will be 0.

15. A rectangular waveguide made of PEC has the following characteristics:

$$b = 1.5\,\text{cm},\ a = 3.0\,\text{cm},\ \mu_r = 1,\ \varepsilon_r = 2.25$$

(a) Calculate the cutoff wavelength and frequency for the TE_{10}, TE_{20}, and TM_{11} modes.

(b) Calculate λ_g (guided wavelength) and (Bonus) Z_0 (mode impedance) at 4.0 GHz.

(c) Calculate the attenuation constant (in dB/cm) at 3.0 GHz. What is the attenuation (in dB) if the guide is 12 cm long? (Hint: Calculate the propagation constant first)

(d) (Bonus) What is the attenuation at frequencies much less than the TE_{10} cutoff frequency?

Hint: Look up what happens to the waves that can't propagate when the frequency of operation is less than the cut-off frequency. Do they get attenuated? Yes! But how?

Solution:

(a) We know, λ_c equals $2a$, a, and $\dfrac{2ab}{\sqrt{a^2+b^2}}$ for the TE_{10}, TE_{20}, and TM_{11} modes, respectively. Thus the cutoff wavelengths are 6.0 cm for TE_{10}, 3.0 cm for TE_{20}, and 2.68 cm for TM_{11}. The cutoff frequencies are a function of the dielectric properties of the insulating material as well as the guide dimensions. The cut-off frequency is given by

$$f_c = \frac{v}{\lambda_c} = \frac{c}{\lambda_c\sqrt{\mu_r\epsilon_r}} = 3.33 \times 10^9 \text{ Hz or 3.33 GHz for the } TE_{10} \text{ mode.}$$

Similarly, $f_c = 6.66$ GHz for the TE_{20} mode and 7.46 GHz for the TM_{11} mode. Assuming $a > b$, the TE_{10} mode always has the lowest cutoff frequency and hence is the dominant mode.

(b) The guide wavelength is obtained from

$$\lambda_g = \frac{\lambda}{\sqrt{1-(f_c/f)^2}} = \frac{\lambda_0}{\sqrt{\mu_r\epsilon_r - (\lambda_0/\lambda_c)^2}}$$

With $\lambda_0 = 7.5$ cm at 4.0 GHz:

$$\lambda_g = \frac{7.5}{\sqrt{2.25 - \left(\dfrac{7.5}{6.0}\right)^2}} = 9.05 \text{ cm}$$

where the TE_{10} mode has been assumed. The characteristic impedance of the guide is calculated from

$$Z_0 = 377\frac{b}{a}\sqrt{\frac{\mu_r}{\epsilon_r}}\frac{\lambda_g}{\lambda}$$

Since $\lambda = \lambda_0/\sqrt{\mu_r\epsilon_r} = 7.5/\sqrt{2.25} = 5.0$ cm,

$$Z_0 = 377 \times \frac{1.5}{3.0} \times \frac{9.05}{\sqrt{2.25} \times 5.0} = 227 \text{ ohms.}$$

Note that a reduction in the guide height b lowers the impedance of the waveguide without changing its TE_{10} cutoff frequency. This is useful when changes in impedance level are required in a waveguide system.

(c) The waveguide is below cutoff to a 3.0 GHz signal and hence the attenuation constant is given by

$$\alpha = \frac{2\pi}{\lambda_c}\sqrt{1-\left(\frac{f}{f_c}\right)^2} \text{ Np/length} = \frac{54.6}{\lambda_c}\sqrt{1-\left(\frac{f}{f_c}\right)^2} \text{ dB/length}$$

With $\lambda_c = 6.0$ cm for the TE_{10} mode:

$$\alpha = \frac{54.6}{6}\sqrt{1-\left(\frac{3.0}{3.33}\right)^2} \approx 3.95 \text{ dB/cm}$$

and

$$\text{total attenuation} = A_\ell = \alpha\ell = (3.95)(12) = 47.4 \text{ dB}.$$

(d) When $f \ll f_c$,

$$\alpha \approx \frac{54.6}{\lambda_c} \text{ dB/length or } A_\ell \approx \frac{54.6}{\lambda_c}\ell \text{ dB}.$$

Therefore,

$$A_\ell = \frac{54.6}{6}(12) = 109.2 \text{ dB}.$$

Note that if the above inequality holds for the dominant mode, then α and A_ℓ are independent of frequency for all modes.